



5D Cluster Density Analysis

Information Density in K-map Minimization

Analysis Date: 2026-01-24

Random Seed: 42

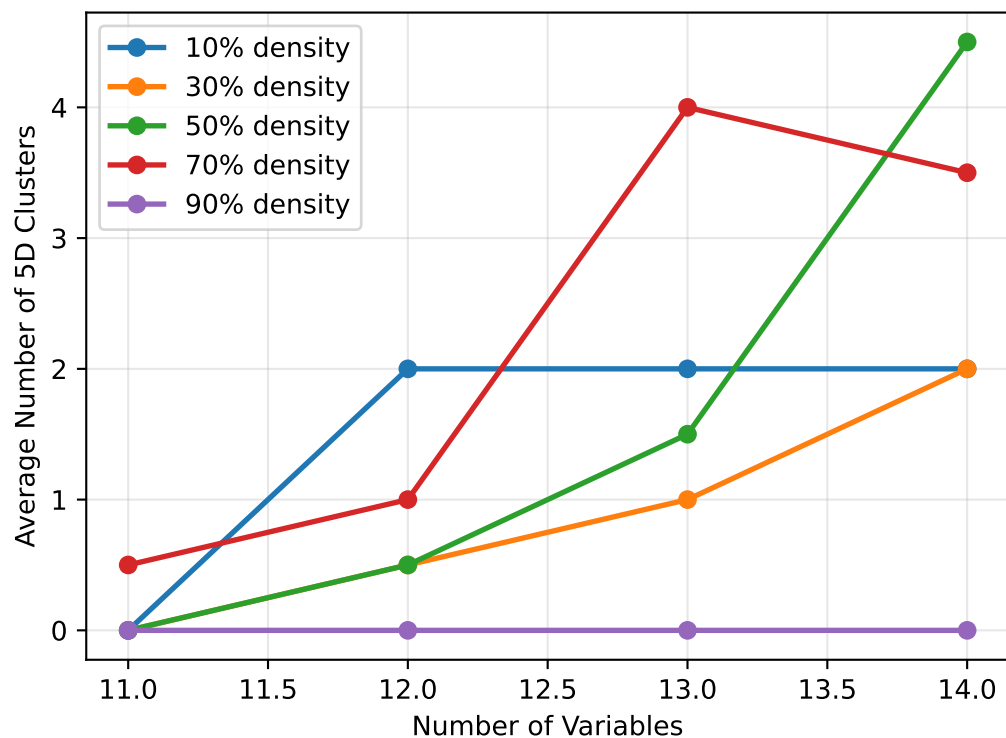
Total Test Cases: 40

Variable Range: 11-14

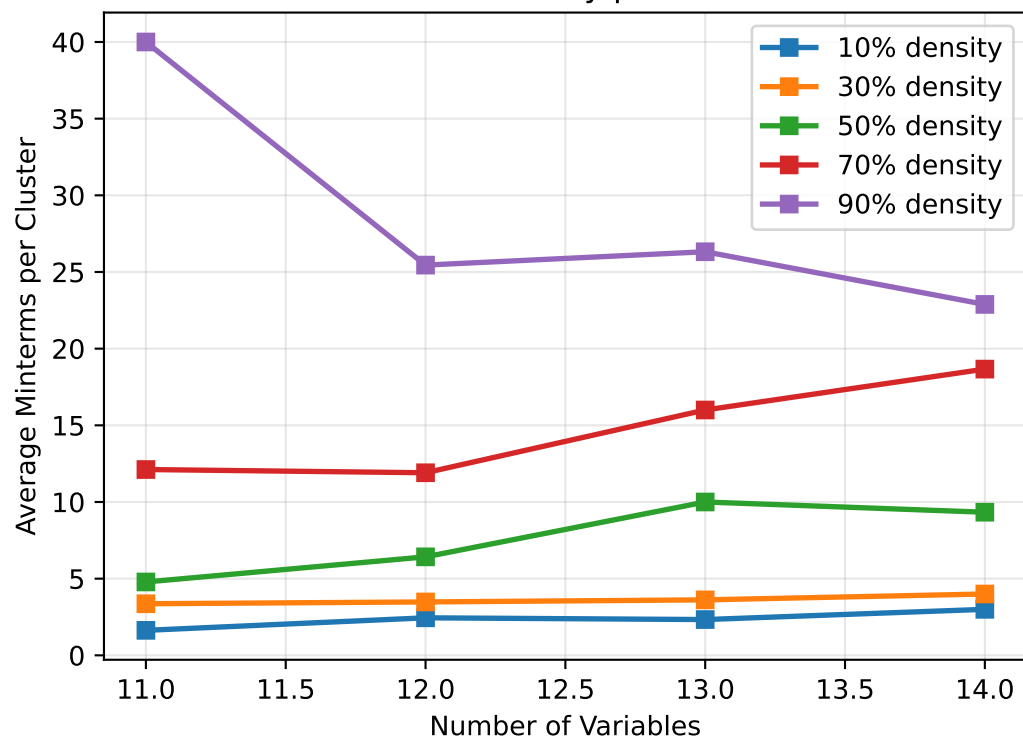
Geometric Clustering Behavior Analysis

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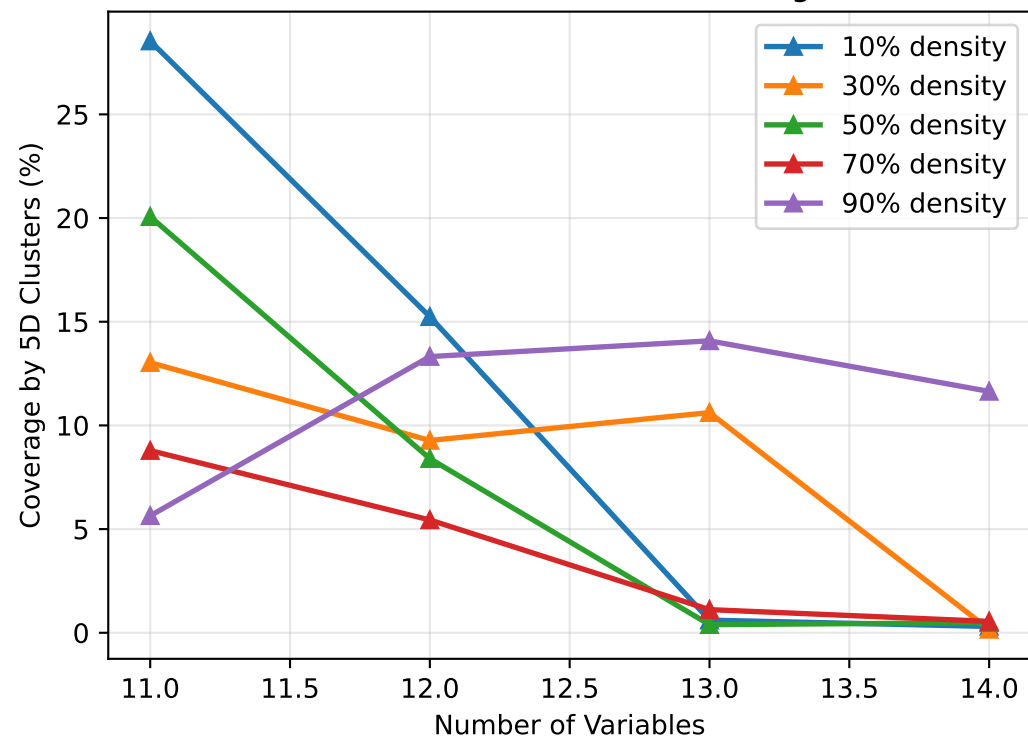
5D Cluster Formation vs Problem Size



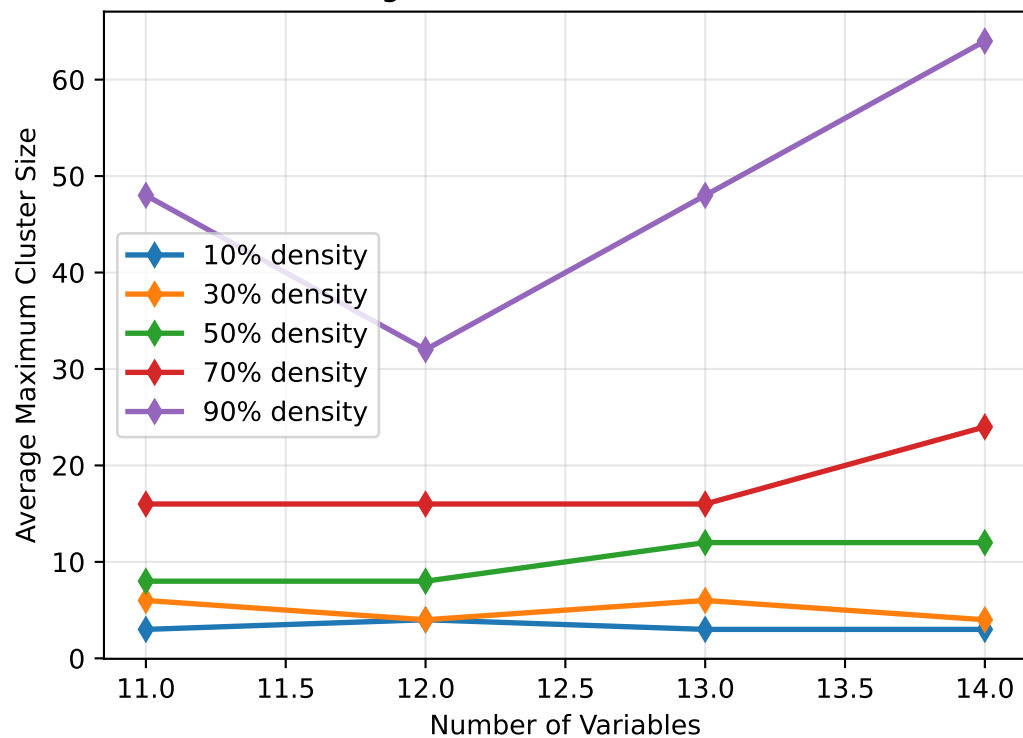
Information Density per 5D Cluster



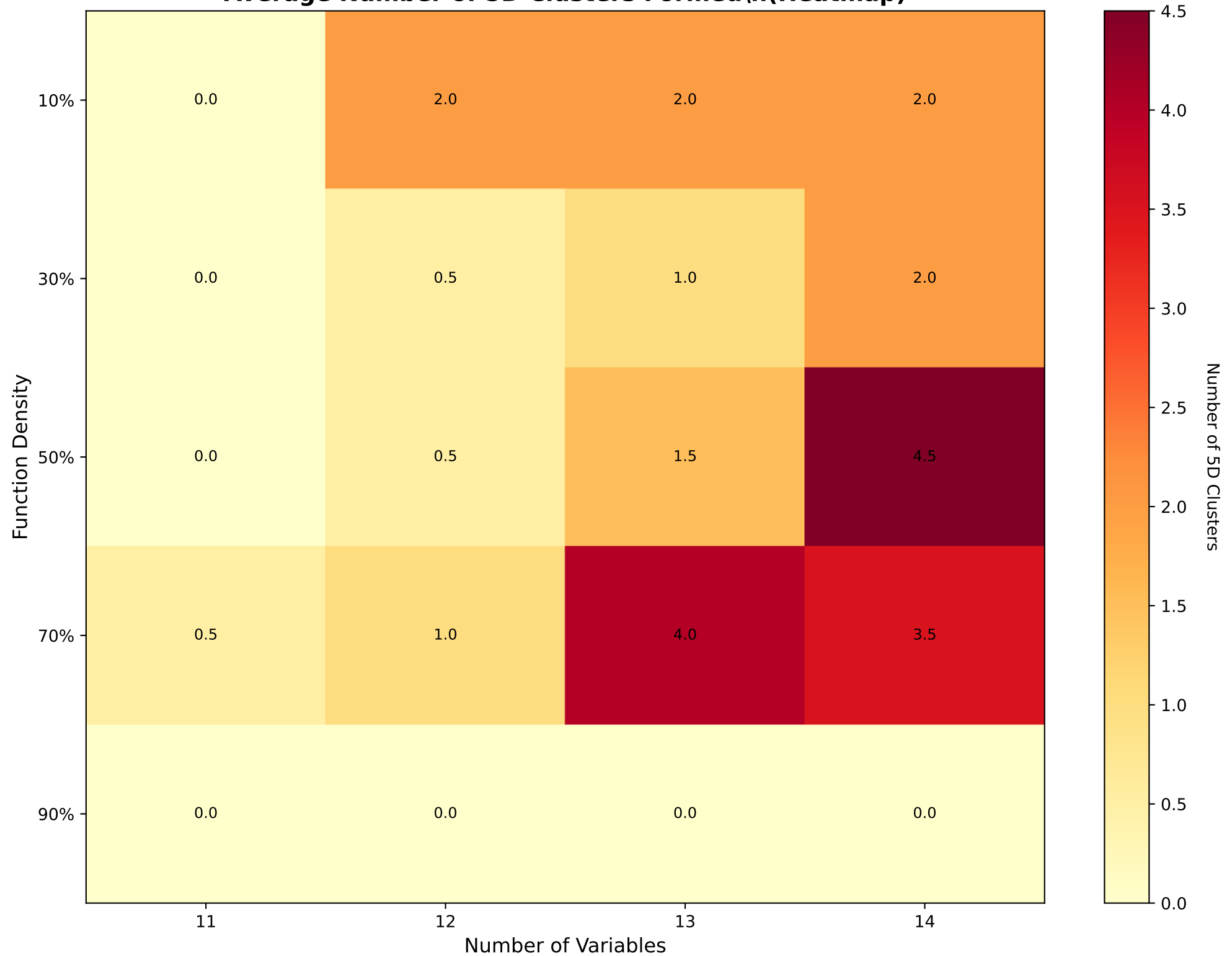
Effectiveness of 5D Clustering

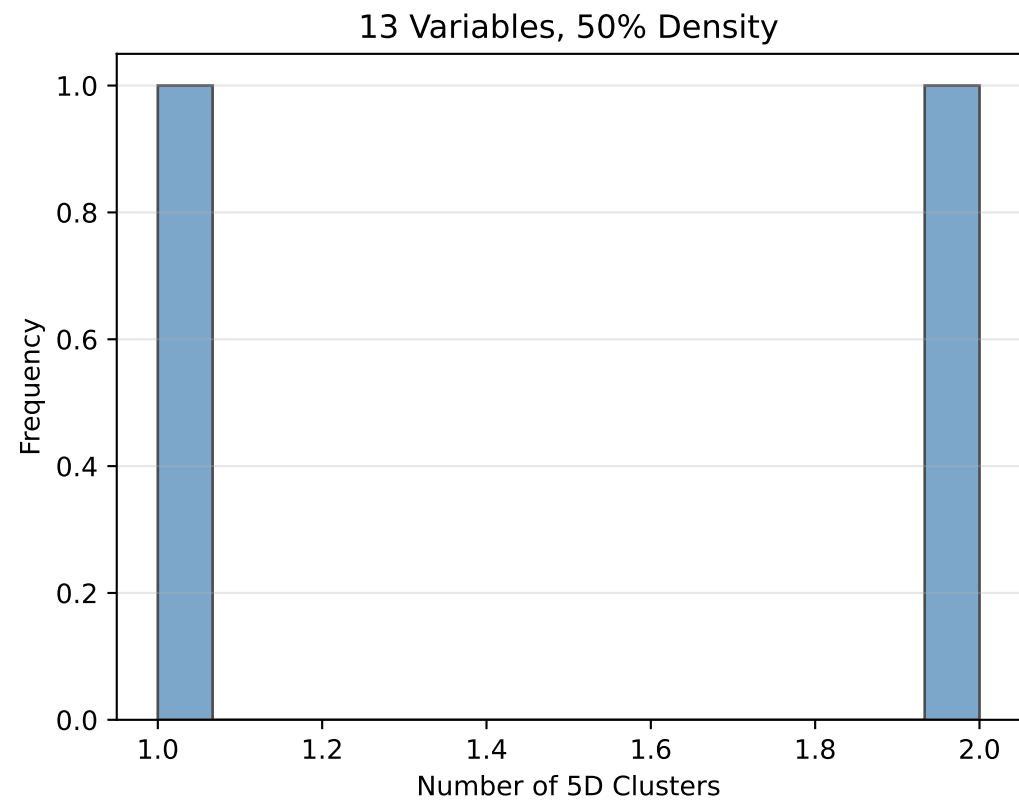
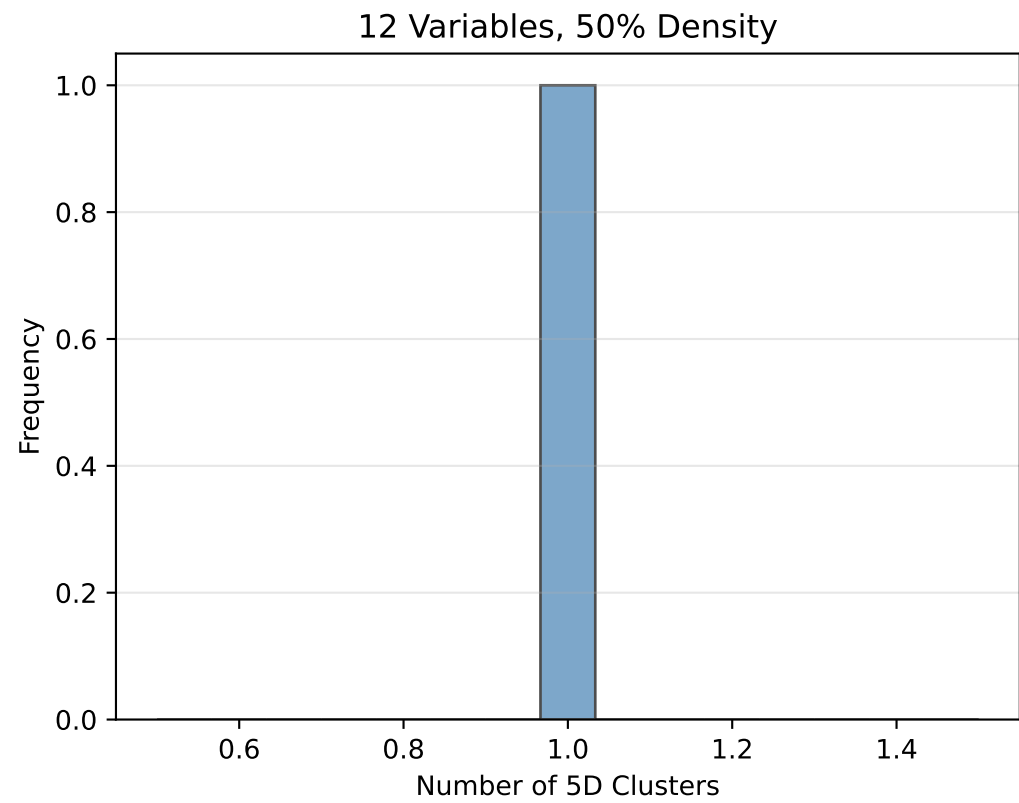
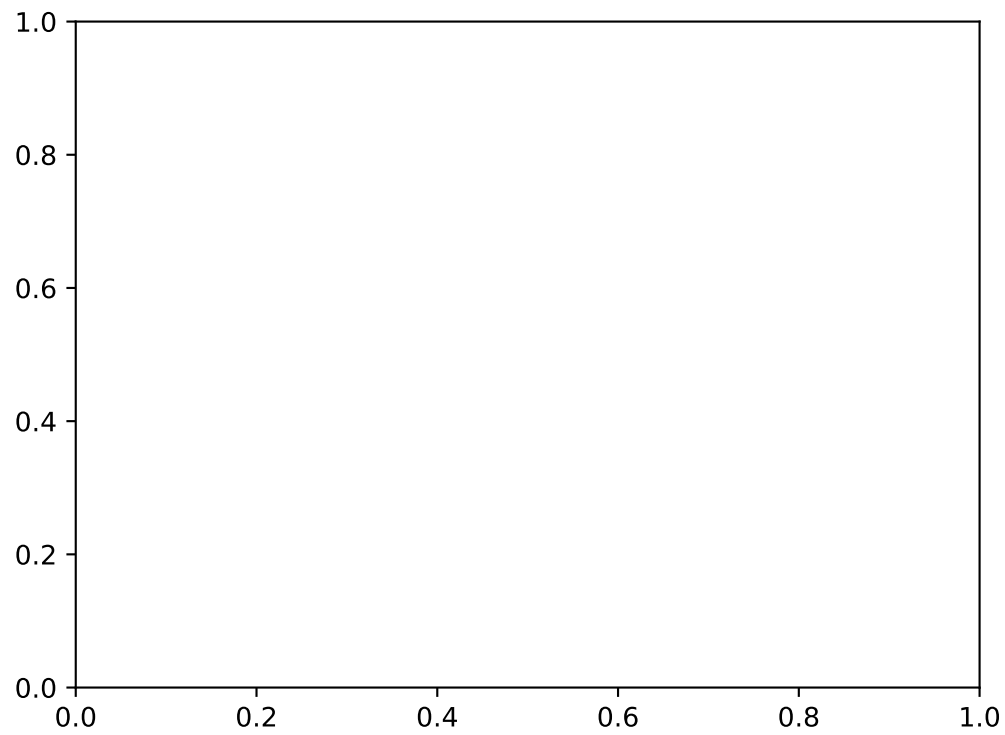


Largest 5D Cluster Formation



Average Number of 5D Clusters Formed\n(Heatmap)





Analysis Summary

Key Findings

- Total tests conducted: 40
- Average 5D clusters per test: 1.25
- Average coverage by 5D clusters: 8.4%
- Average hyperspan: 2.00 hyperchunks
- Variable range: 11 to 14
- Density levels tested: 5

Observations

- *5D cluster formation increases with variable count ($n > 10$)*
- *Higher density functions tend to form more 5D clusters*
- *Hyperspan increases with problem complexity*
- *5D clusters span multiple 10-variable hyperchunks*
- *Coverage by 5D clusters validates hierarchical approach*